

Finley Bartholmai

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Education

Worcester Polytechnic Institute

BS, Electrical and Computer Engineering — Class of 2029

Worcester, MA

Cumulative GPA: 4.0/4.0 | Presidential Scholarship | 2025 Fall & 2026 Spring Semester Dean's List

Relevant coursework:

- Microelectronic Circuits I & II
- Introduction To RF Circuit Design
- Software Radio Design - designed digital PLLs, filters, and demodulators in Python
- Continuous-Time Signal and System Analysis
- Operating Systems - Low-level systems programming in C/Rust

Technical Skills

- **Programming Languages:** Python, C, C++, Rust, C#, MATLAB, Java, JavaScript, Verilog (HDL)
- **Design & Simulation:** LTSpice, OnShape (CAD&FEA), Fusion 360, MasterCAM, Ansys Maxwell/HFSS, FEMM, SimScale
- **Software:** Git, STM32 Toolchain, ESP-IDF, pico-sdk, Foxglove, Desmos, Typst, SymPy, Z3 Prover, FreeRTOS
- **Miscellaneous:** Soldering (SMD & rework), CNC manufacturing, 3D printing, Linux, SDRs, Using test equipment, FMEA

Projects

BLDC Motor Controller(s) (https://www.tinkerneering.com/projects/bldc_v2.html)

May 2025 — Present

- Designed a ~1/2kW brushless motor drive using a custom FOC implementation for high bandwidth current control
- Created thermal and electrical models to characterize transient power handling capacity and switching characteristics.

Differential Slider-Arm (https://www.tinkerneering.com/projects/diffy_slider.html)

Apr 2025 — Jun 2025

- Used MATLAB for system identification and tuning of LQG control scheme
- Lightweight 3D printed coaxial planetary gearbox

RISC-V SoC (<https://github.com/NifleySnifley/CPUDesign-IS>)

Aug 2024 — Jan 2025

- Implemented the rv32im instruction set using Verilog to synthesize with the yosys toolchain
- Simulation of HDL designs with Verilator, Icarus Verilog, and cocotb; testbenches in SystemVerilog
- Pipelined core capable of 33.4 MIPS @ 50MHz with flash bootloader on Lattice ECP5 LFE5U-25F

Rocket Tracker (<https://www.tinkerneering.com/projects/rockettracker.html>)

Apr 2023 — Aug 2024

- Designed hardware, firmware, and software for a custom amateur rocketry tracking system (KiCAD, ESP-IDF, PyQt)
- Created a flexible data link protocol for streaming telemetry information in real time to custom ground station software
- Developed FreeRTOS based firmware for high frequency data logging and fault-tolerant control of the tracker

Experience

Raytheon (RTX) — Intern, Receiver-Exciter & Processing Architecture Dept.

May 2026 — July 2026

- Setup validation & CI pipeline for test software (MATLAB)
- Signal processing and data analysis.

WPI Formula SAE — Electrical team member & Driverless steering actuation lead

Aug 2025 — Present

- PCB design, firmware development, real-time signal processing algorithms for custom magnetic sensors
- Telemetry, log analysis, troubleshooting

WPI Wireless Association — Historian

Aug 2025 — Present

- Amateur Extra class ham radio operator. Participant in contesting, foxhunts and archival QSL card digitization.

WPI High Power Rocketry Club — Electronics & Programming sub-team member

Aug 2025 — May 2026

- Rocket antenna design lead — responsible for telemetry and video antennas for the club's two rockets

FIRST Robotics Competition Team 2530 "Inconceivable"

Aug 2019 — Jun 2025

- Team captain, robot design/CAD lead (2024-25).
- Helped lead the team to the FRC World Championship in 2024 and 2025